

WASP-11b

R-1577

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-10_131018 · created 2026-08-01T23:23:55+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2456579.98105 +/- 0.00023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1765 +/- 0.0021
Transit depth (Rp/Rs)^2	3.12 +/- 0.075 [%]
Orbital Inclination (inc)	89.28 +/- 0.34
Airmass coefficient 1 (a1)	0.41 +/- 0.13
Airmass coefficient 2 (a2)	0.3404 +/- 0.0056
Scatter in the residuals of the lightcurve fit is	29.52 %
Best Comparison Star	#3 - [630, 386]
Optimal Aperture	5.43
Optimal Annulus	14.14
Transit Duration (day)	0.01204 +/- 0.0005

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1765 $\pm$ 0.0021 vs 0.131 (deviation 21.67 σ)
Duration fit vs published	0.01204 d vs 0.1075 d (deviation 190.92 σ)
Mid-time O-C	-5358.4 min
Depth SNR	84.0
Residual scatter	29.52 %

Light curve

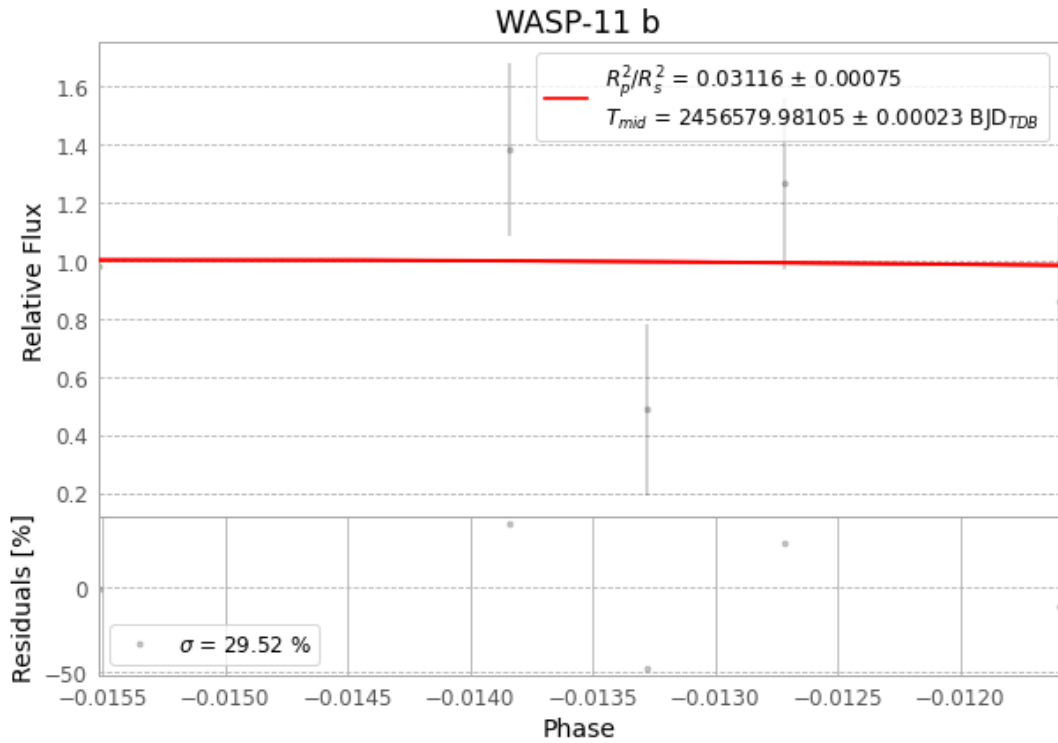


Figure 1 — FinalLightCurve_WASP-11 b_2013-10-17.png (SHA-256 aaad80ae1451c432...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [568, 400], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3be2fba7cdae2102...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

75 FITS frames (49 MB), HATP-10131018032112.FITS → HATP-10131018070312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-131018.log (ee2bc1e173eb...), FinalLightCurve_WASP-11 b_2013-10-17.png (aaad80ae1451...), FinalLightCurve_WASP-11 b_2013-10-17.csv (e7cf3b73f110...)

Compute resources

Wall time 155.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 23:23Z.